# Chapter 9

# Global properties of continuous functions

Up to now we studied *local* properties of functions (that is, properties of functions as $x \to x_0$, where $x_0$ can be a finite point or $\pm\infty$). In the current chapter we study *global* properties of functions. That is, we consider functions defined on intervals $[a,b] \subset \mathbb{R}$ (where $a < b$) and analyze their behavior on this entire interval.

**Continuity on a closed interval**

Throughout this chapter, "$f$ is continuous on $[a,b]$" means that $f$ is continuous at every point of $(a,b)$, right-continuous at $a$ and left-continuous at $b$.

Theorem 6.15(7a) still holds at the endpoints for sequences in $[a,b]$: if $x_n \in [a,b]$ for all $n$ and $\lim_{n\to\infty} x_n = a$, then $\lim_{n\to\infty} f(x_n) = f(a)$, and similarly at $b$ (only points of $[a,b]$ are involved, so one-sided continuity suffices).

## 9.1 Existence of zeros: Bolzano's Theorem

A **zero** of a function $f$ is a point $x_0$ such that $f(x_0) = 0$. Our first result guarantees the existence of zeros:

**Theorem 9.1** (Bolzano's Theorem)**:** Let $f : \mathrm{dom}(f) \subseteq \mathbb{R} \to \mathbb{R}$ and suppose it is continuous on a *closed* interval $[a,b] \subseteq \mathrm{dom}(f)$ where $a < b$. If $f(a)$ and $f(b)$ have opposite signs, i.e. $f(a)f(b) < 0$, then there exists $x_0 \in (a,b)$ such that $f(x_0) = 0$. Moreover, if $f$ is *strictly* monotone on $[a,b]$ then $x_0$ is the *unique* point in $[a,b]$ where $f$ equals zero.

*Proof.* Assume that $f(a) < 0 < f(b)$. The proof for the case $f(a) > 0 > f(b)$ will follow the exact same ideas. We split the proof into several steps. The ideas are essentially identical to the *bisection method* that we've seen in the proof of the Bolzano-Weierstrass Theorem (see §5.3)

**Step A.** Constructing sequences $\{a_n\}_{n\in\mathbb{N}}$, $\{b_n\}_{n\in\mathbb{N}}$ in search of $x_0$:

**Step A0.** Define $a_0 = a$ and $b_0 = b$. Define the center point between $a_0$ and $b_0$ to be $c_0 = \frac{a_0+b_0}{2}$. If $f(c_0) = 0$ then $x_0 = c_0$ and the proof is done. Otherwise, if $f(c_0) < 0$ we define $a_1 = c_0$ and $b_1 = b_0$ and if $f(c_0) > 0$ we define $a_1 = a_0$ and $b_1 = c_0$.

**Step A1.** Define the center point between $a_1$ and $b_1$ to be $c_1 = \frac{a_1+b_1}{2}$. If $f(c_1) = 0$ then $x_0 = c_1$ and the proof is done. Otherwise, if $f(c_1) < 0$ we define $a_2 = c_1$ and $b_2 = b_1$ and if $f(c_1) > 0$ we define $a_2 = a_1$ and $b_2 = c_1$.

**Step A2.** Define the center point between $a_2$ and $b_2$ to be $c_2 = \frac{a_2+b_2}{2}$. If $f(c_2) = 0$ then $x_0 = c_2$ and the proof is done. Otherwise, if $f(c_2) < 0$ we define $a_3 = c_2$ and $b_3 = b_2$ and if $f(c_2) > 0$ we define $a_3 = a_2$ and $b_3 = c_2$.

...and so on...

**Step A$n$.** The points $a_n$ and $b_n$ have been defined in the previous step. Define the center point between $a_n$ and $b_n$ to be $c_n = \frac{a_n+b_n}{2}$. If $f(c_n) = 0$ then $x_0 = c_n$ and the proof is done. Otherwise, if $f(c_n) < 0$ we define $a_{n+1} = c_n$ and $b_{n+1} = b_n$ and if $f(c_n) > 0$ we define $a_{n+1} = a_n$ and $b_{n+1} = c_n$.

...and so on...

**Step B.** Properties of the sequences $\{a_n\}_{n\in\mathbb{N}}$, $\{b_n\}_{n\in\mathbb{N}}$.

If a zero $x_0$ hasn't been located in any finite step, then by induction we have constructed sequences $\{a_n\}_{n\in\mathbb{N}}$, $\{b_n\}_{n\in\mathbb{N}}$ satisfying:

$$\begin{aligned}
&a_0 \le a_1 \le \cdots \le a_n \le \cdots \le b_n \le \cdots \le b_1 \le b_0\\
&[a_0, b_0] \supset [a_1, b_1] \supset \cdots \supset [a_n, b_n] \supset \cdots\\
&f(a_n) < 0 < f(b_n) \quad \text{and} \quad b_n - a_n = \frac{b_0 - a_0}{2^n}, \quad \forall n \in \mathbb{N}.
\end{aligned}$$

By the first property, $\{a_n\}_{n\in\mathbb{N}}$ is a bounded, monotonically increasing sequence (it is bounded by $b_0$, for instance). Hence, by the Monotone Convergence Theorem (Theorem 5.1), its limit exists; denote it by $a_\infty \in \mathbb{R}$:

$$\lim_{n\to\infty} a_n = a_\infty.$$

Similarly, $\{b_n\}_{n\in\mathbb{N}}$ is bounded and monotonically decreasing, hence its limit exists by the same theorem. We denote it by $b_\infty \in \mathbb{R}$:

$$\lim_{n\to\infty} b_n = b_\infty.$$

**Step C.** Conclusion.

Observe that

$$b_\infty - a_\infty = \lim_{n\to\infty} b_n - \lim_{n\to\infty} a_n = \lim_{n\to\infty} (b_n - a_n) = \lim_{n\to\infty} \frac{b_0 - a_0}{2^n} = 0$$

so that $b_\infty = a_\infty$. Denote this number by $x_0$:

$$x_0 = b_\infty = a_\infty.$$

Since $a_n \in [a_0, b_0] = [a, b]$ for all $n$, Proposition 5.3 gives $x_0 \in [a, b]$. Since $f$ is continuous, we may use Theorem 6.15(7a) (the substitution theorem for a continuous function of a convergent sequence) to get:

$$\lim_{n\to\infty} f(a_n) = \lim_{n\to\infty} f(b_n) = f(x_0).$$

However we know that

$$f(a_n) < 0 < f(b_n), \quad \forall n \in \mathbb{N}.$$

By Theorem 6.15(4),

$$f(x_0) = \lim_{n\to\infty} f(a_n) \leq 0 \quad \text{and} \quad 0 \leq \lim_{n\to\infty} f(b_n) = f(x_0)$$

Hence it necessarily follows that

$$f(x_0) = 0.$$

Since $f(a) < 0 < f(b)$, $x_0 \neq a$ and $x_0 \neq b$, hence $x_0 \in (a, b)$.

Finally, if $f$ is strictly monotone on $[a, b]$, then by Proposition 4.1, $f$ is injective on $[a, b]$. This means that for every $y \in f([a, b])$ there exists a unique $x \in [a, b]$ such that $f(x) = y$. Hence $x_0$ is the unique zero of $f$ in $[a, b]$. □

> **The word "algorithm"**
>
> *The proof above is also a practical way of computing a zero of $f$, known as the* bisection method *(we already used the same idea in the proof of the Bolzano-Weierstrass Theorem, Section 5.3, keeping each time the half that contains infinitely many terms of the sequence). After $n$ steps we know that $x_0$ lies in $[a_n, b_n]$, an interval of length $\frac{b-a}{2^n}$, so the midpoint $c_n$ approximates $x_0$ with an error of at most $\frac{b-a}{2^{n+1}}$. Every step halves the error: starting from an interval of length 1, twenty steps bring it below $10^{-6}$ (since $2^{20} > 10^6$). Other methods are faster, but bisection cannot fail as long as $f$ is continuous and changes sign, and it is still used in numerical software, often combined with faster methods.*
>
> *A step-by-step procedure of this kind is called an* algorithm. *The word comes from Muhammad ibn Musa al-Khwarizmi, a scholar at the House of Wisdom in Baghdad, the academy founded in the 9th century by Caliph al-Ma'mun, where Greek scientific works were translated into Arabic. Al-Khwarizmi wrote a treatise explaining the decimal place-value system developed in India (where Brahmagupta had already given rules for computing with zero in the 7th century). Its Latin translation opens with* Dixit Algoritmi *("al-Khwarizmi says");* algorismus *came to mean computing with these numerals, and was later respelled "algorithm" by confusion with the Greek* arithmos *("number"). His book on solving equations,* al-jabr*, gave us "algebra." This is also why the digits we use are called "Arabic numerals" in Europe.*

**Example 9.1:** The function

$$f(x) = \begin{cases} 1 & x \geq 10 \\ -1 & x < 10 \end{cases}$$

changes sign on the interval $[9, 11]$. However, there is no zero in $[9, 11]$ (i.e. there is no $x \in [9, 11]$ such that $f(x) = 0$). Why? The problem with this $f$ is that it is not continuous on $[9, 11]$ (there's a jump discontinuity at $x = 10$).

**Example 9.2:** The function $f(x) = x^2$ is a continuous function on $\mathbb{R}$ that has a zero at $x = 0$, however it is always non-negative (i.e. it does not change sign). This shows that a continuous function can have a zero without changing sign. That is, changing sign is a *sufficient* condition for a continuous function to have a zero, but not a *necessary* condition.

**Example 9.3:** The function $f(x) = e^x + \sin x$ is continuous on $\mathbb{R}$. Let's look at the interval $[-\frac{\pi}{2}, 0]$. For $x = -\frac{\pi}{2}$ the function is negative: $f(-\frac{\pi}{2}) = e^{-\pi/2} + \sin(-\frac{\pi}{2}) = e^{-\pi/2} - 1 < e^0 - 1 = 1 - 1 = 0$ and for $x = 0$ the function is positive: $f(0) = e^0 + \sin 0 = 1 + 0 = 1 > 0$. Hence there exists $x_0 \in (-\frac{\pi}{2}, 0)$ such that $f(x_0) = 0$. Moreover, $e^x$ is strictly increasing on $\mathbb{R}$, and $\sin x$ is strictly increasing on $[-\frac{\pi}{2}, 0]$, so that $x_0$ is the unique zero within this interval.

**Corollary 9.2:** Let $(\alpha, \beta) \subseteq \mathbb{R}$ be an interval ($\alpha$ may be $-\infty$ and $\beta$ may be $+\infty$), and let $f : (\alpha, \beta) \to \mathbb{R}$ be continuous and satisfy

$$\lim_{x \to \alpha^+} f(x) = \ell_\alpha \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$$
$$\lim_{x \to \beta^-} f(x) = \ell_\beta \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$$

where $\ell_\alpha < 0 < \ell_\beta$ or $\ell_\alpha > 0 > \ell_\beta$. Then $f$ has a zero $x_0$ in $(\alpha, \beta)$: $f(x_0) = 0$. Moreover, if $f$ is strictly monotone in $(\alpha, \beta)$ then $x_0$ is unique.

*Proof.* This proof is an exercise. *Hint:* by the Local Sign Theorem (Theorem 6.2, applied to one-sided limits), there are $a < b$ in $(\alpha, \beta)$ such that $f(a)$ and $f(b)$ have opposite signs; now apply Theorem 9.1 on $[a, b]$. □

Continuity cannot be dropped: the step function in the example above, on the interval $(9, 11)$, has limits $-1$ at $9^+$ and $1$ at $11^-$, but no zero.

**Corollary 9.3:** Let $f, g : \mathbb{R} \to \mathbb{R}$ be continuous on an interval $[a, b]$. If

$$f(a) < g(a) \quad \text{and} \quad f(b) > g(b)$$
or
$$f(a) > g(a) \quad \text{and} \quad f(b) < g(b)$$

then there exists a point $x_0 \in (a, b)$ satisfying

$$f(x_0) = g(x_0).$$

Moreover, if one of $f$, $g$ is strictly increasing and the other is decreasing (or one is strictly decreasing and the other increasing), then $x_0$ is unique.

*Proof.* This proof is very simple: we consider the function

$$h(x) = f(x) - g(x).$$

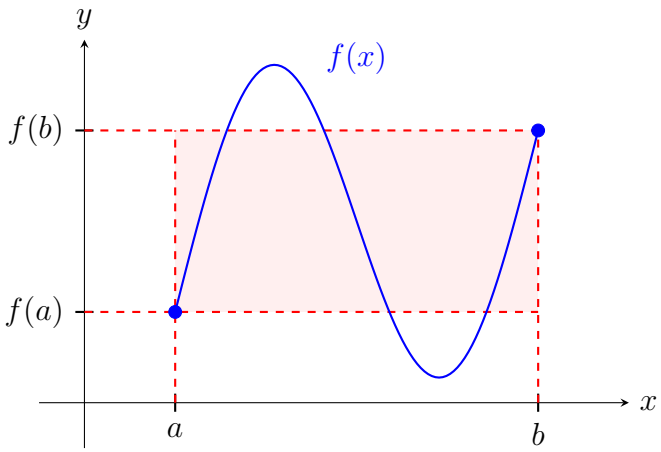
Then $h$ is continuous on $[a, b]$ (since it is the difference of two continuous functions on $[a, b]$). Furthermore, $h(a)$ and $h(b)$ have different sign (due to the assumptions on $f$ and $g$). So $h$ satisfies the conditions of Theorem 9.1, and there exists $x_0 \in (a, b)$ such that $h(x_0) = 0$. But this means (by definition of $h$) that $f(x_0) = g(x_0)$.

If $f$ is strictly increasing and $g$ is decreasing, then $-g$ is increasing, so $h = f + (-g)$ is strictly increasing by Lemma 4.2, and the uniqueness part of Theorem 9.1 applies. The other case is similar. □

It is not enough that $f$ and $g$ are both strictly increasing: $f(x) = x^3$ and $g(x) = x$ satisfy all the hypotheses on $[-2, 2]$, but $f = g$ at $x = -1, 0, 1$.

## 9.2 Range of a continuous function defined on an interval

**Theorem 9.4** (Intermediate Value Theorem)**:** Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on the closed interval $[a, b]$, where $a < b$. Then $f$ attains all values between $f(a)$ and $f(b)$: for every $y_0$ between $f(a)$ and $f(b)$ there exists $x_0 \in [a, b]$ such that $f(x_0) = y_0$.




*Proof.* If $f(a) = f(b)$ the statement of the theorem is trivial: $[f(a), f(b)]$ is a single point, and it has both $a$ and $b$ in its pre-image. So we assume that $f(a) < f(b)$. The case $f(a) > f(b)$ follows similarly, and is left as an exercise.

We need to prove that for any $f(a) < y_0 < f(b)$ there exists $a < x_0 < b$ such that $f(x_0) = y_0$. Consider the constant function $g(x) = y_0$ for all $x \in [a, b]$. Then

$$f(a) < y_0 = g(a) \quad \text{and} \quad g(b) = y_0 < f(b).$$

Thus $f(a) < g(a)$ and $f(b) > g(b)$. By Corollary 9.3 there is a point $x_0 \in (a, b)$ where $f(x_0) = g(x_0)$. By the definition of $g$, we have that $f(x_0) = y_0$ and the proof is complete. □

**Corollary 9.5:** Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on an interval $I \subseteq \mathbb{R}$ (open, closed or half-open, bounded or unbounded). Then $f(I)$ is also an interval, with endpoints given by $\inf_I f$ and $\sup_I f$.

*Proof.* If $f(I)$ is a single point there's nothing to prove. So assume that there exist $y_1 \neq y_2$ both belonging to $f(I)$, and assume, without loss of generality, that $y_1 < y_2$. Then there exist $x_1 \neq x_2$, both elements of $I$, such that $f(x_1) = y_1$ and $f(x_2) = y_2$. Assume that $x_1 < x_2$ (the other case will follow a similar proof). Then by the Intermediate Value Theorem (Theorem 9.4), $f$ attains all values $y \in [y_1, y_2]$:

$$[y_1, y_2] \subseteq f([x_1, x_2]) \subseteq f(I).$$

Therefore we have shown that for any two points in $f(I)$, the closed interval that has these points as end-points is a subset of $f(I)$. This implies that $f(I)$ is itself an interval (possibly infinite). This fact is discussed in Lemma 2.5 (Chapter 2). By the definition of *infimum* and *supremum* of a set, it follows that the endpoints of $f(I)$ are $\inf_I f$ and $\sup_I f$ (they can be infinite). □